PHY4605 Computational Methods in Physics
Learning Note | Week 9

Data Fitting: Infer a physical parameter and read residuals with uncertainty

Use a small physical dataset to estimate a parameter, inspect where the model misses the data, and report only the uncertainty the evidence supports.

Physical question. A resistor is labelled 47 Ω, but measured voltage and current are not perfectly exact. What resistance is supported by the full dataset, how can residuals reveal model mismatch, and how precise should the reported answer be?

Learning outcomes

1. fit a supplied simple physical relationship and interpret each fitted parameter with its unit;
2. calculate and interpret residuals to judge whether a model misses visible structure; and
3. validate a fitted result and state uncertainty cautiously using supplied repeat measurements.

Uncertainty labels. A repeatability range is the observed span of results from repeated runs. It is not a manufacturer tolerance, which is an allowed component specification, or a confidence interval, which is a statistical interval from a stated inference procedure.

How to use this note. Follow the physical question to the equation, trace the plain-language steps, then read the line-numbered MATLAB starter code. Pause at each prediction and validation check; the Guided extension and Optional extension are not required for the ordinary route.

1. Measurements are not the same as model parameters

The experiment supplies pairs of current I and voltage V. These are **measured quantities**. We propose the simple relationship

$$V = RI + b,$$

where R is the resistance and b is a small voltage offset. The values of R and b are **fitted parameters**: they are inferred from the pattern across all measurements.

The units follow directly from the equation. Because voltage is in volts and current is in amperes,

$$[R] = \frac{\mathrm{V}}{\mathrm{A}} = \Omega, \qquad [b] = \mathrm{V}.$$

The unit check is already useful: if a reported slope were labelled only in volts, its physical interpretation would be incomplete.

Prediction checkpoint. For a resistor near 47 Ω, increasing current by 0.01 A should increase voltage by about 0.47 V. The voltage–current graph should therefore have a positive slope near 47 V/A.

2. Start with the data, then choose the fit

The supplied teaching dataset is small enough to inspect directly.

I (A)	V (V)	I (A)	V (V)
0.01	0.48	0.05	2.36
0.02	0.93	0.06	2.83
0.03	1.43	0.07	3.28
0.04	1.86	0.08	3.77

Plotting before fitting matters. A command can always return coefficients, but a visibly curved trend, one extreme point, or a unit mistake may make the proposed straight-line model inappropriate.

3. A fitting algorithm in plain language

For this week, the least-squares calculation itself is supplied. Least squares chooses the line that makes the total of squared residuals as small as possible. The required reasoning is the sequence around it:

1. plot the measured data with labelled axes and units;
2. state the proposed relationship and identify the parameter units;
3. fit a straight line to all measured pairs;
4. evaluate the fitted model at the measured current values;
5. calculate residual = measured voltage − fitted voltage;
6. inspect whether the residuals scatter around zero or show a pattern;
7. validate the fitted resistance against independent physical evidence; and
8. report only the precision supported by the repeatability evidence.

4. MATLAB starter code for the fit

Base MATLAB provides `polyfit` and `polyval`. A degree of 1 requests a straight line. The first coefficient is the slope and the second is the intercept. The starter code below calculates the fitted values and residuals.

```
I_A = (0.01:0.01:0.08)';
V_measured_V = [0.48 0.93 1.43 1.86 ...
                2.36 2.83 3.28 3.77]';

p = polyfit(I_A,V_measured_V,1);
R_fit_ohm = p(1);
offset_fit_V = p(2);

V_fit_V = polyval(p,I_A);
residual_V = V_measured_V - V_fit_V;
```

Listing 1: MATLAB fit and residual calculation for the supplied voltage–current data

The fitted values are

$$R = 47.0000\,\Omega, \qquad b = 0.0025\,\mathrm{V}.$$

The slope is the physically interesting resistance. The small fitted intercept can be read as a possible voltage zero offset in this simple measurement model.

Do not confuse a fitted parameter with a measured value. No row in the table "measures $R = 47\,\Omega$" directly. The resistance estimate comes from the change of voltage with current across the dataset.

5. Overlay the model and measurements

The fitted line follows the overall trend while individual measurements sit slightly above or below it.

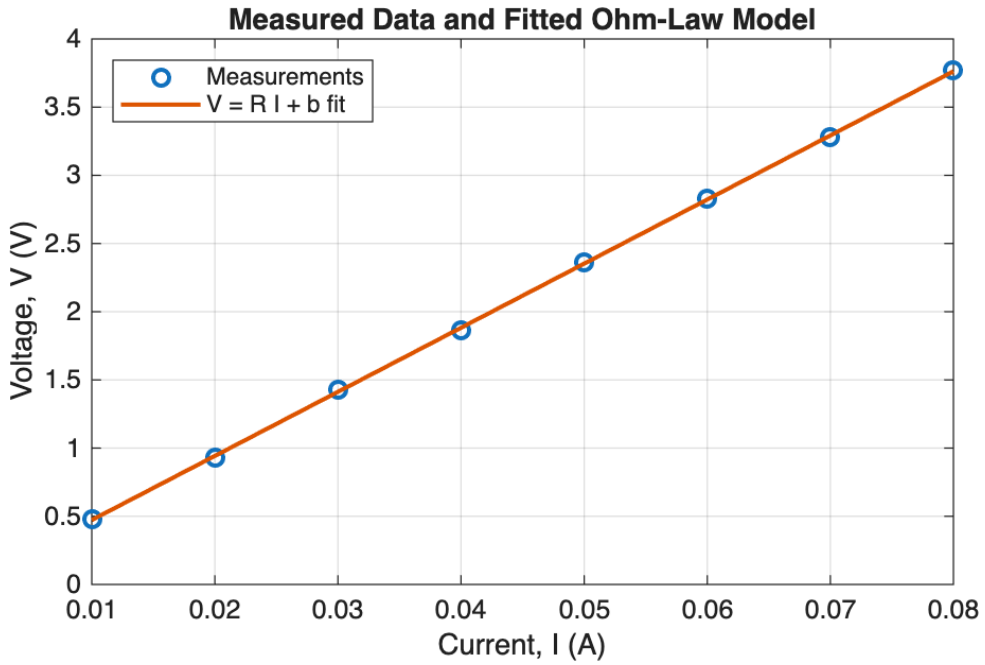


Figure 1: Supplied MATLAB figure: measured voltage–current points and the fitted $V = RI + b$ model. The slope is $47.0000\,\Omega$.

An overlay is useful, but visual closeness alone is not enough. The next step asks where and by how much the model misses each measurement.

6. Residuals make model–data disagreement visible

For measurement i, define

$$r_i = V_{i,\text{measured}} - V_{i,\text{fit}}.$$

A positive residual means the measured voltage is above the fitted line. A negative residual means it is below. Residuals have unit volts because they are voltage differences.

For the supplied data, the residual root-mean-square value is about

$$\text{RMSE}_r = 0.0130\,\text{V},$$

and the largest absolute residual is $0.0225\,\text{V}$.

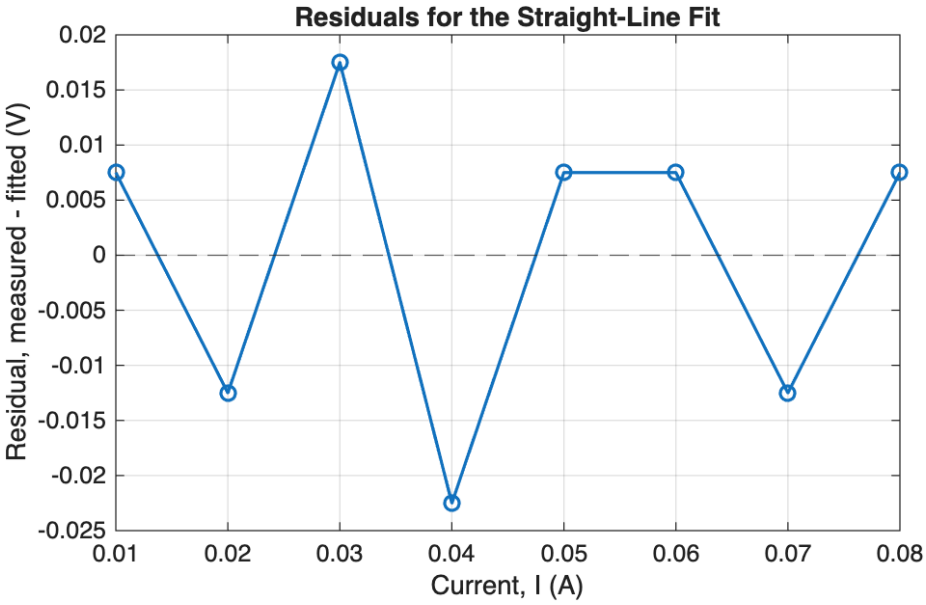


Figure 2: Residuals alternate around zero and show no obvious curved pattern in this small dataset.

The absence of obvious structure supports the straight-line model *over this measured range*. It does not prove that the resistor is perfectly ohmic at every current or temperature.

Required model-quality check. Small residuals are not enough by themselves. Ask whether they show a pattern. A U-shaped sequence—positive at both ends and negative in the middle—would suggest missing curvature or another systematic effect.

7. Validate the fitted resistance with a physical reference

The resistor is labelled 47 Ω with a nominal 5% tolerance. The manufacturer range is therefore

$$47(1 - 0.05) = 44.65\,\Omega, \qquad 47(1 + 0.05) = 49.35\,\Omega.$$

The fitted value 47.0000 Ω lies inside this range. This is an independent physical/reference check because the nominal component information did not come from the fitting calculation.

Tolerance is not fit uncertainty. The 5% label describes the component specification. It does not tell us how much the fitted resistance would change if we repeated this measurement procedure. Keep these two ideas separate.

8. State uncertainty only as strongly as the evidence allows

Three supplied repeat runs give fitted slopes

$$47.00, \quad 47.06, \quad 47.27\,\Omega.$$

For Week 9, use the observed span as a simple repeatability range:

$$47.0 \text{ to } 47.3\,\Omega.$$

A defensible statement is:

Cautious result. The fitted resistance is about 47.1 Ω, with repeat runs spanning roughly 47.0–47.3 Ω for this supplied measurement procedure. This range is not a formal confidence interval and does not include every possible systematic error.

Notice the reporting precision. Writing $47.111111\,\Omega$ would imply more certainty than the repeated data support.

9. Worked reasoning chain

Use the same chain whenever a familiar physical dataset is fitted:

Step	Week 9 evidence
Physical model	$V = RI + b$ for an approximately ohmic resistor
Variables and units	I in A, V and b in V, R in Ω
Algorithm	plot $\rightarrow$ fit $\rightarrow$ predict $\rightarrow$ residuals $\rightarrow$ validate
Code	supplied `polyfit` and `polyval` starter code
Error/uncertainty	residual scatter and repeat-run range
Validation	fitted R inside the independent nominal component range
Interpretation	linear model is adequate over the supplied current range

This sequence is more important than memorising a fitting function name. In an assessment, the syntax may be supplied; you still need to explain the physical parameter, unit, residual evidence, and justified conclusion.

Guided extension: one fit-quality statistic

This guided extension asks you to read supplied fit evidence and interpret its meaning; independent reproduction of the statistic is not required. You may be shown a supplied coefficient of determination, R^2. Values closer to 1 indicate that the fitted predictions account for more of the observed variation in that dataset. Treat it as supporting evidence only. A high R^2 does not repair wrong units, a poor physical model, a structured residual pattern, or an unverified data-entry error.

You may also be shown two plausible simple models and their residual evidence. Compare the supplied evidence; independent reproduction of advanced selection metrics is not required.

Optional extension

Weighted fitting, nonlinear fitting, covariance matrices, regularisation, and formal confidence intervals are outside the Week 9 required route. These topics are not required for ordinary pass-level work here.

Three takeaways

1. A fitted parameter must keep the unit implied by the physical equation.
2. Residuals show where the model misses the measurements; visible structure can reveal a poor model even when the overlay looks convincing.
3. Report uncertainty only to the level supported by the evidence, and keep repeatability distinct from manufacturer tolerance or formal confidence intervals.

Preparation for the practical. Be ready to fit supplied straight-line starter code, state units, read residual structure, use one independent check, and report only justified precision.